

Hugo Folloni Guarilha

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Professional Summary

Full-stack software engineer with 4+ years of experience building robust, scalable applications. Expertise in tackling complex data challenges using React, .NET, Python, and Docker to develop critical systems for educational platforms, government institutions, and technology companies. Backed by a Computer Science degree from UFRJ and C2 professional English proficiency.

Skills

JavaScript | TypeScript | React | C# | .NET | Python | Django | Node | Express | Redux | Next.js | PostgreSQL | SQL Server | MongoDB | Git
Frontend | Backend | Full-Stack | SQL | NoSQL | AWS | Docker | Scrum | Agile | English and Portuguese – *Professional proficiency or above*

Experience

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|--|--------------------------------|-------------------------------|--------------------------|
| Jr. Software Engineer | <u>BigDataCorp</u> | <i>Rio de Janeiro, Brazil</i> | 01/2026 - present |
| <ul style="list-style-type: none">Enabled consolidated financial reporting for leadership by orchestrating the end-to-end development of core billing platforms (React, .NET), streamlining complex data flows for contracts, invoices, and accounts receivable.Engineered the transition to a unified Price Tables model, successfully migrating data for over 5,000 enterprise clients and 10,000 contracts to support 9 product lines and scale the billing ecosystem.Implemented a proactive caching layer that reduced data retrieval times by up to 90%, and achieved 100% automated test coverage for the core API by establishing internal TDD protocols, using xUnit. | | | |
| Software Engineer Intern | <u>BigDataCorp</u> | <i>Rio de Janeiro, Brazil</i> | 06/2024 - 12/2025 |
| <ul style="list-style-type: none">Architected key features for internal administrative panels, delivering RESTful and event-driven APIs (.NET 8) and responsive (React) components to enable critical financial and selling data management and streamline workflows for 50+ internal users.Built and deployed automated Purchase Order (PO) tracking systems and credit workflows, replacing manual Omie ERP checks and directly accelerating daily operations for the Finance and Sales teams, integrating AWS S3 document saving.Spearheaded a major stack modernization effort, upgrading legacy backend APIs to .NET 8 and migrating essential User Interface components to a React 18 architecture. | | | |
| Scientific Research Fellow - ETL | <u>COPPETEC</u> | <i>Rio de Janeiro, Brazil</i> | 06/2023 - 12/2025 |
| <ul style="list-style-type: none">Designed and deployed 10+ new Talend ETL workflows and maintained a larger 40-job data integration architecture, securely processing over 1.67B+ rows of legal and institutional data across 600+ database tables using PostgreSQL and Docker.Executed zero-downtime database operations and targeted data corrections across production PostgreSQL and SQL Server environments for a major public institution, strictly maintaining data integrity and operational reliability for critical state systems.Pioneered solutions for workspace sharing, creating a method to host the Talend application in Docker containers. | | | |
| Scientific Research Fellow - Development | <u>Machine Teaching</u> | <i>Rio de Janeiro, Brazil</i> | 02/2022 - 12/2025 |
| <ul style="list-style-type: none">Directed the build of a synchronous assessment module on a major Computer Science educational platform as my capstone thesis project, unifying the platform's evaluation cycle within a secure, time-controlled exam environment.Optimized user data dashboards by rewriting database queries, reducing load times by over 70%.Automated backend and User Interface testing suites with Playwright and Django Testing, establishing a strong foundation for platform reliability. | | | |

Education

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|--|--|-------------------------------|--------------------------|
| Bachelor of Science | <u>Universidade Federal do Rio de Janeiro</u> | <i>Rio de Janeiro, Brazil</i> | 02/2021 - 11/2025 |
| <ul style="list-style-type: none">Major in Computer Science (Top-ranked university in Brazil, recognized for excellence in STEM fields)
Capstone Thesis: "Implementation of Synchronous Assessments on the Machine Teaching Platform" | | | |

Projects

- vinyl-scrobbler**: Engineered a full-stack application (**React 19**, **TypeScript**, **.NET 9**) to digitize physical vinyl playback, integrating the Alexa Skills Kit for voice-activated scrobbling and utilizing Spotify/Last.fm APIs for automated metadata retrieval and timestamp calculations.
- yze.bio**: A full-stack 'link-in-bio' creator with customizable layouts, built with **React**, **C#**, **.NET**, and **PostgreSQL**.
- spotify-multiverse**: A music discovery app (**React 18**, **Python**) that generates personalized recommendations by applying Linear Algebra to playlist attributes.

Freelancer

- **Instituto de Proteção ao Jogador (2025):** Developed the complete system (**React, Node.js, PostgreSQL**) for a psychological project aimed at helping individuals with gambling addiction, designed to support a broader betting platform with 700k+ users.
- **Bracell - Floresta Sempre Viva (2022):** Architected, coded, and maintained an interactive environmental web application (React) in partnership with a major enterprise, independently managing all phases of development and deployment.

Publications & Presentations

- **Author** – WEI 2026 (Brazilian Computer Society Congress): *"Beyond Correctness: A Competency-Driven Framework for Designing Autograder Test Suites"* (DOI: 10.5753/wei.2026.21827).
- **Co-author** — Revista Brasileira de Informática na Educação (RBIE), 2025: "Technical and Tactical Factors of an Educational Platform for a Large Number of Students" (DOI: 10.5753/rbie.2025.4479) - Award-Winning Article.
- **Presenter** — Semana de Integração Acadêmica (SIAC UFRJ), 2024: Received **Best Presentation Award** for the study "Testes de Servidor no Machine Teaching"
- **Presenter** — Semana de Integração Acadêmica (SIAC UFRJ), 2025: Received **Best Presentation Award** for the study "Funcionalidade de Avaliação Síncrona na Plataforma Machine Teaching"

Others

- **Certificate of English Language Proficiency C2 (EF SET)** – The highest-level certificate of English proficiency (**02/2025**)